Project- Microsoft Sentinel

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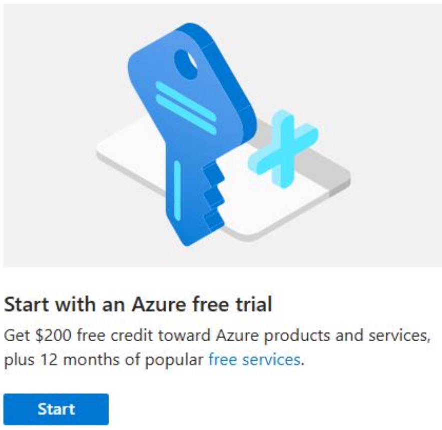
Introduction

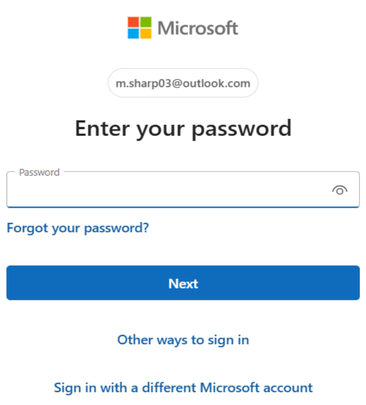
Microsoft Azure Sentinel is a cloud-native Security Information and Event Management (SIEM) and Security Orchestration, Automation, and Response (SOAR) solution. It's designed to help Security Operations Center (SOC) analysts detect, investigate, and respond to security threats across an enterprise's digital environment.

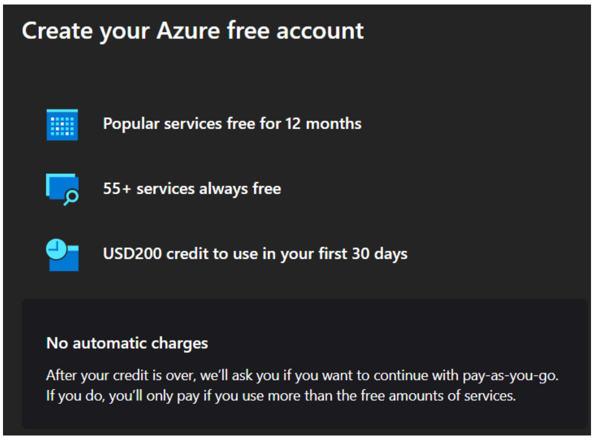
Step 1: Set up a free Azure Account

To create a free account, visit https://azure.microsoft.com/en-gb/pricing/purchase-options/azure-account and click the Sign-Up button in the Azure Free Account section. You will get a $200 USD free trial for 30 days.



You will then be prompted to enter your username and password. If you do not have one, you will need to create one. You will then enter your profile details and add a phone number which will trigger a verification code to be sent to you. Add the code, then click next. You will need to add a credit or debit card for identity verification. However, you will not be charged if you decide to move to pay-as-you go pricing.

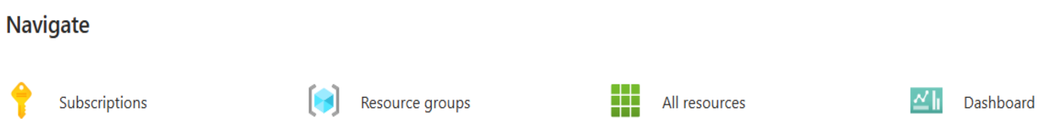




Step 2: Create a Resource Group

A **resource group** is a container that holds related resources for an Azure solution. You will first need to create a resource group to organize your resources.

1. **Log in to the Azure Portal**:
   - Go to portal.azure.com and log in with your Azure account.
2. **Navigate to Resource Groups**:
   - In the left-hand menu, click on **Resource groups**.

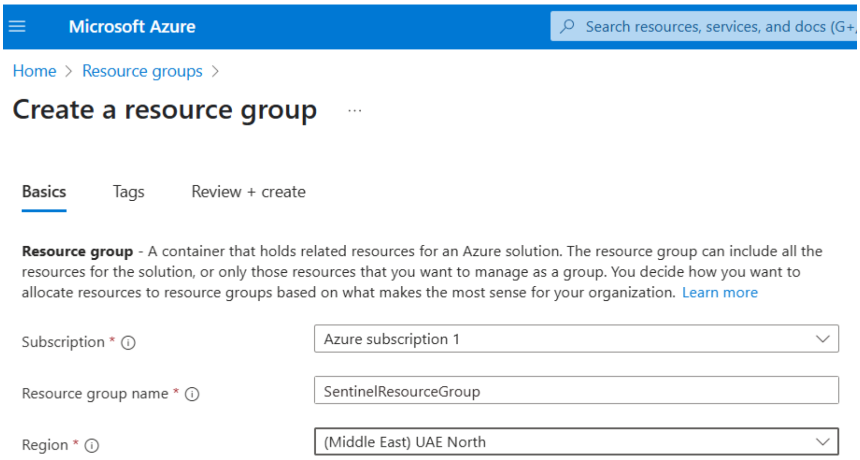


Click the blue button to create a resource group.

3. **Create a New Resource Group**:
   - Click on **+ Create** at the bottom.

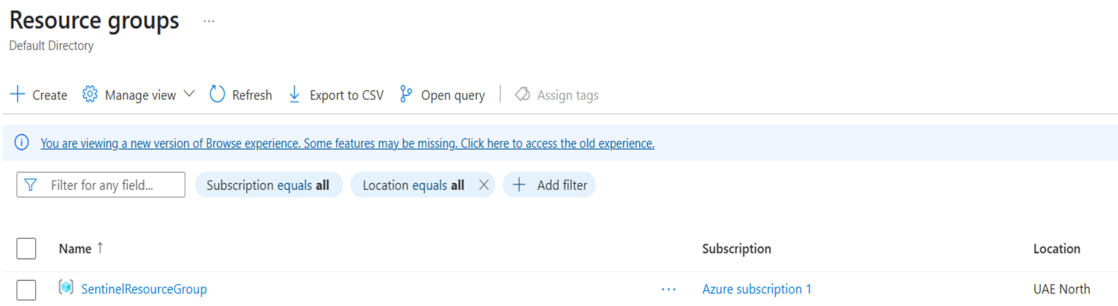
   - **Subscription**: Select the subscription you want to use.

- **Resource Group Name**: Give your resource group a name, such as *SentinelResourceGroup*.
- **Region**: Select the region closest to your location or based on your organization's requirements.



- Click **Review + Create**, then **Create** after validation.

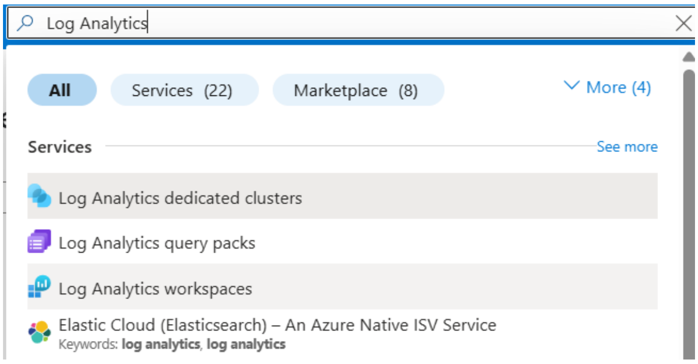
When you refresh the page, you should see the resource page that you just created.



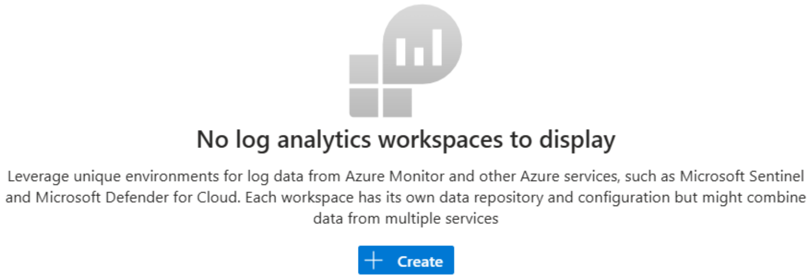
Step 3: Create a Log Analytics Workspace

The **Log Analytics Workspace** is where Azure Sentinel will store its logs and data for analysis.

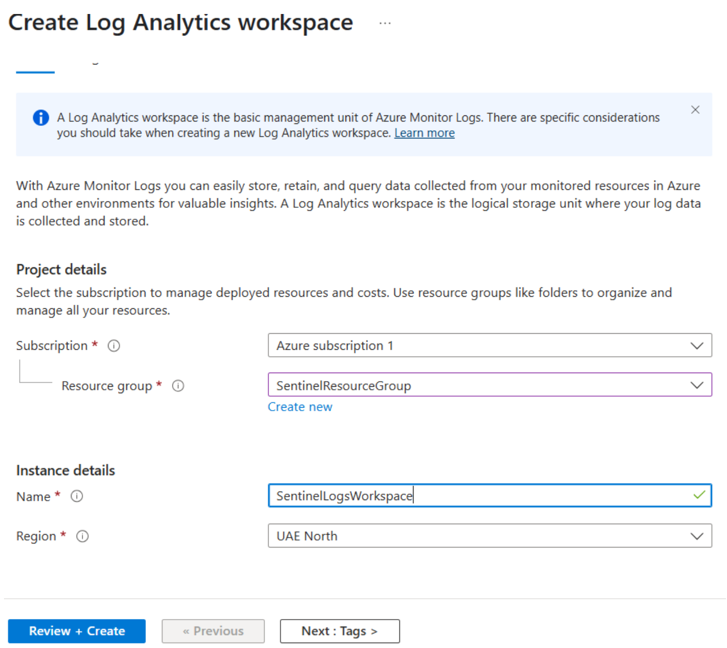
1. **Navigate to Log Analytics Workspaces**:
   - In the search bar at the top, type "Log Analytics" and select **Log Analytics Workspaces**.



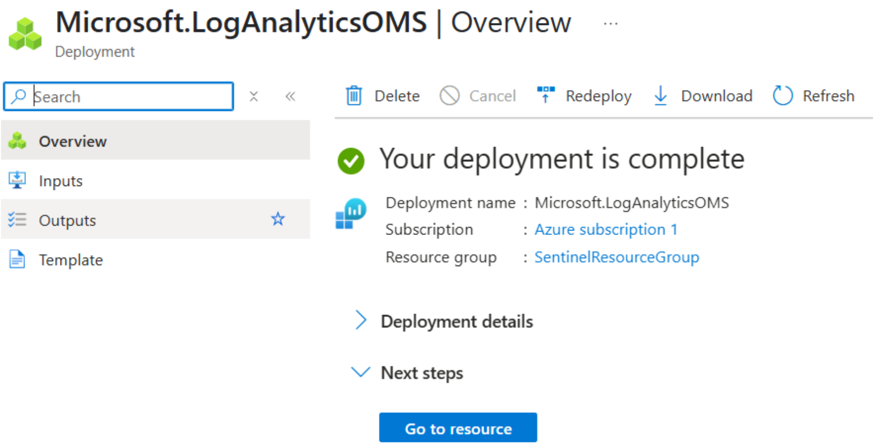
2. **Create a New Log Analytics Workspace**:
   - Click **+ Create**.



   - **Subscription**: Choose your subscription.
   - **Resource Group**: Select the resource group you created in Step 2.
   - **Name**: Provide a name for your workspace, such as *SentinelLogsWorkspace*.
   - **Region**: Choose the same region as your resource group.
   - Click **Review + Create**, then **Create** after validation.



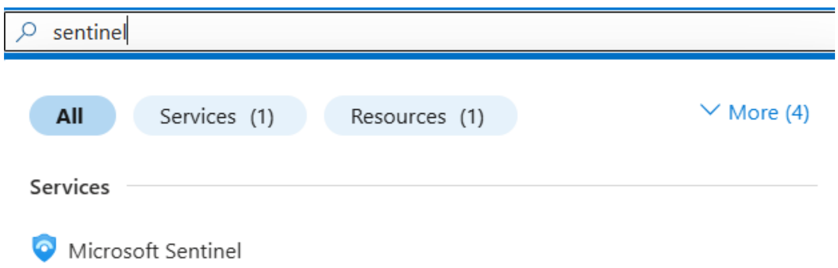
After you have successfully deployed the workspace, the following notification will appear.



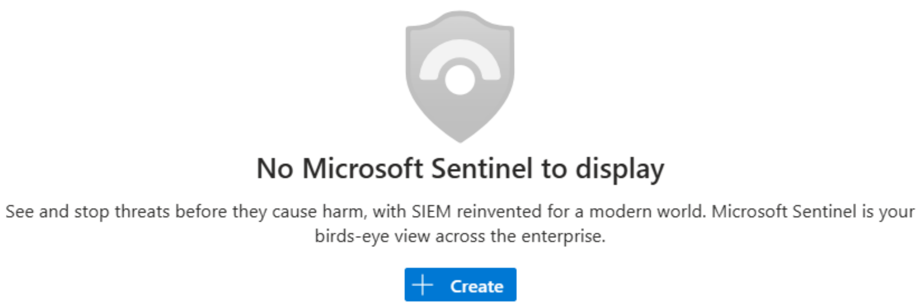
Step 4: Create Microsoft Sentinel

Microsoft Sentinel provides SIEM (Security Information and Event Management) capabilities, allowing you to monitor security across your infrastructure.

1. **Navigate to Microsoft Sentinel**:
   - In the search bar, type "Microsoft Sentinel" and select it.

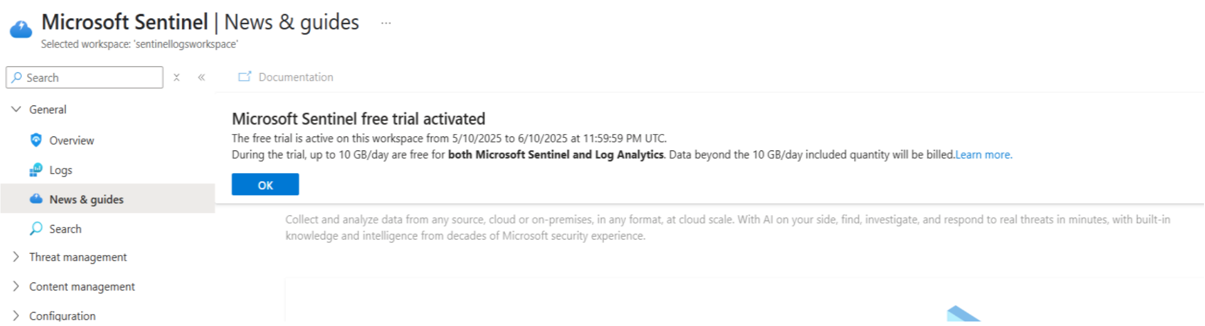


2. **Add Microsoft Sentinel to Your Workspace**:
   - Click **+ Add** in the Microsoft Sentinel pane. Click the blue button to create a Microsoft Sentinel.



- **Select Workspace**: Choose the Log Analytics Workspace you created in Step 2 (SentinelLogsWorkspace).
- Click **Add** to attach Sentinel to your workspace.

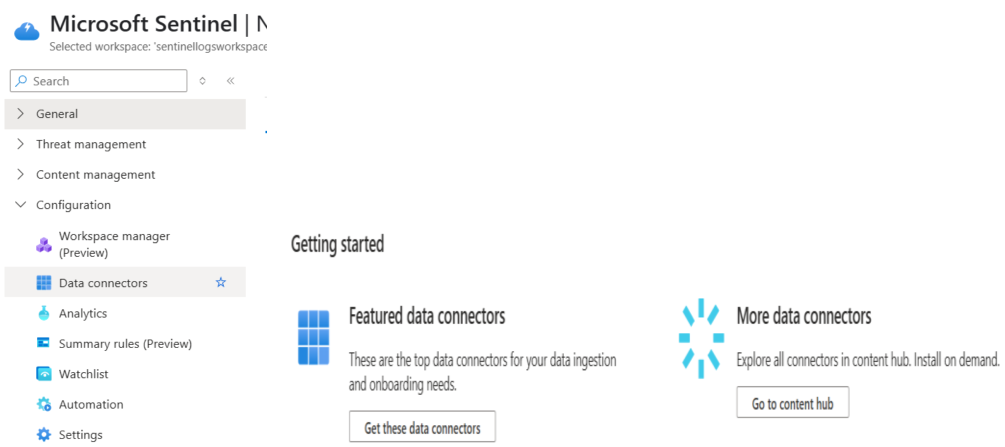
You will see it is added and activated with the free subscription trial for 30 days.



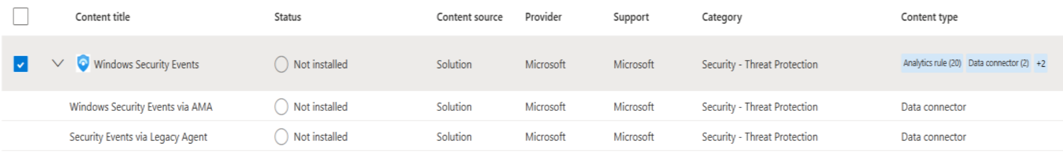
Step 5: Install the Windows Security Events Content Pack

Content packs provide pre-built templates, queries, and dashboards to help you monitor specific windows security events.

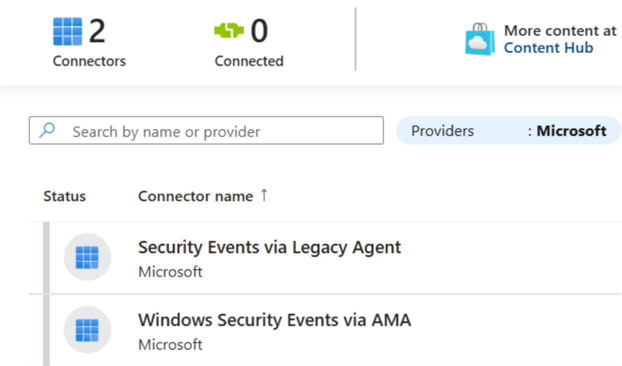
1. **Open Microsoft Sentinel**:
   - Go to **Microsoft Sentinel** from the portal.
2. **Add Data Connector**:
   - In the **Microsoft Sentinel** left panel, click on **Data connectors** from the left-hand menu. Next, scroll to the bottom of the page and click on the 'Go to content hub' option.



In the search bar type in '**Windows Security Events**' and choose the **Windows Security Events** option. In the lower right corner of the page, click the blue install button.



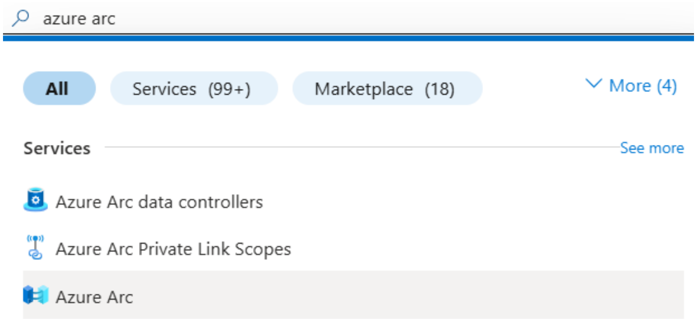
Refresh and go back to the data connectors page, then you will see the two connectors.



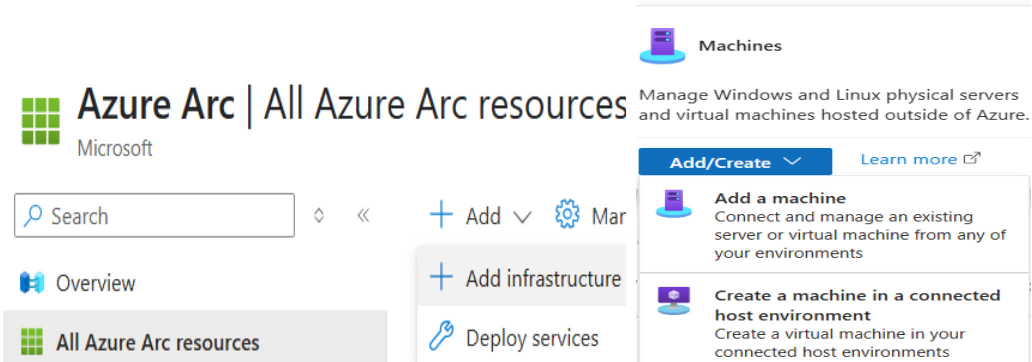
Step 6: Install Azure Arc on a Device

Azure Arc allows you to manage on-premise or multi-cloud resources in Azure.

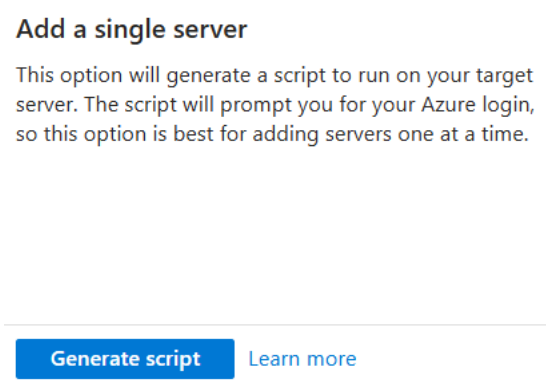
1. **Navigate to Azure Arc**:
   - In the search bar, type **Azure Arc** and select **Azure Arc**.



In the left pane, choose 'All Azure Arc resources', followed by the add button, choosing 'Add infrastructure' from the dropdown menu. Click on the 'Add/Create' button in the Machines section and click 'Add a machine'.

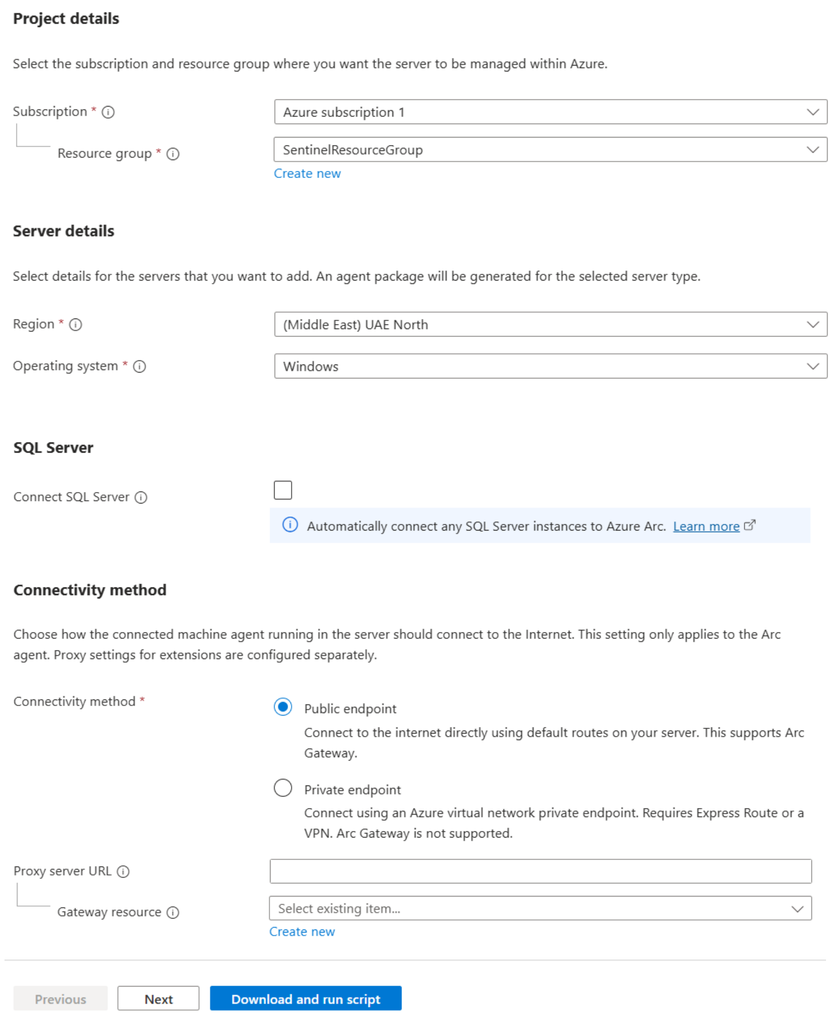


2. **Add a Server**:
   - Click on **+ Add** in the Azure Arc pane.
   - Choose **Servers** as the resource type you want to connect. Click on a "add a single server" window.

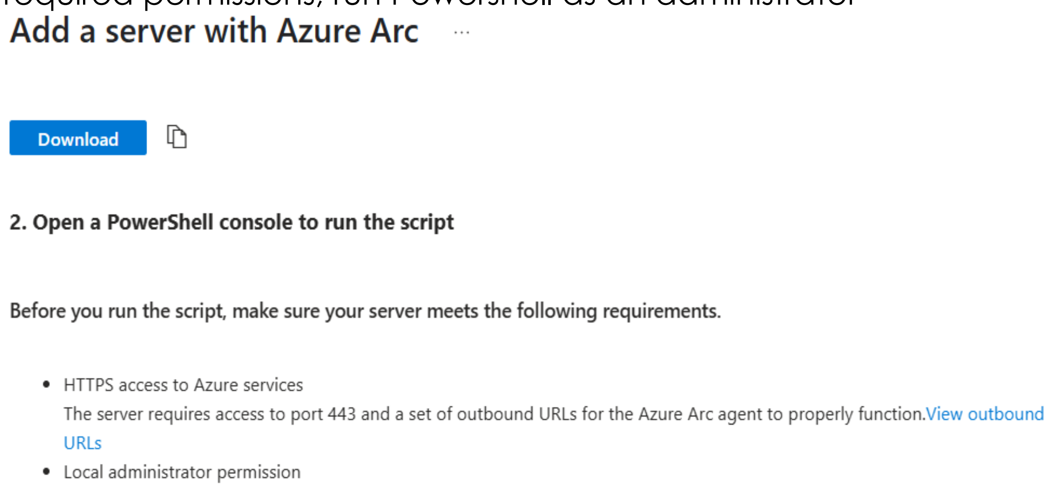


3. **Generate the Script to Install Azure Arc**:
   - Choose your **Subscription** and **Resource Group** (*SentinelResourceGroup*).
   - Provide a **Region** for the resource.
   - **Download the script** provided by Azure Arc.

- Run the script on the target device (e.g., a Linux or Windows server) to install and register Azure Arc.



On the next page download the created script to your device. Once downloaded, type in the ‘powershell’ in the search bar on your PC. To have all the required permissions, run Powershell as an administrator



Change your directories to the downloads folder using the

command: *cd $HOME\Downloads*

```
PS C:\WINDOWS\system32> cd "$HOME\Downloads"
PS C:\Users\Mallikah\Downloads> cd $Home\Downloads
```

Run the script you have downloaded on your device. You may have to temporarily enable script execution.

Run the command: *Set-ExecutionPolicy -ExecutionPolicy RemoteSigned -Scope Process*

```
PS C:\Users\Mallikah\Downloads> Set-ExecutionPolicy -ExecutionPolicy RemoteSigned -Scope Process

Execution Policy Change
The execution policy helps protect you from scripts that you do not trust. Changing the execution policy might expose
you to the security risks described in the about_Execution_Policies help topic at
https:/go.microsoft.com/fwlink/?LinkID=135170. Do you want to change the execution policy?
[Y] Yes  [A] Yes to All  [N] No  [L] No to All  [S] Suspend  [?] Help (default is "N"):
```

You might be prompted to confirm if you trust this script. Type 'Y' and proceed using the command below.

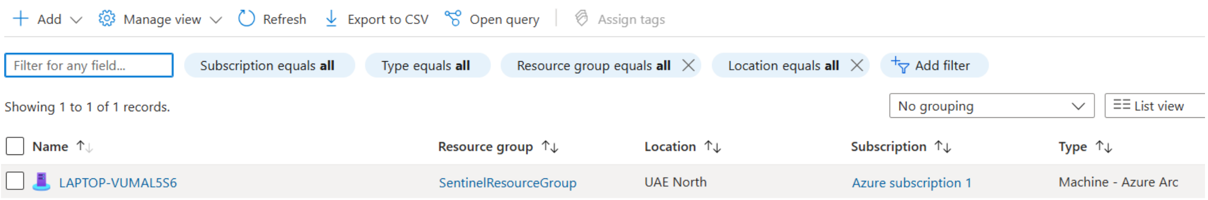
Command: .*OnboardingScript.ps1*

```
PS C:\Users\Mallikah\Downloads> .\OnboardingScript.ps1
VERBOSE: Installing Azure Connected Machine Agent
VERBOSE: PowerShell version: 5.1.26100.3624
VERBOSE: Total Physical Memory: 8192 MB
VERBOSE: .NET Framework version: 4.8.9032
VERBOSE: Checking if this is an Azure virtual machine
VERBOSE: Error Unable to connect to the remote server checking if we are in Azure
VERBOSE: Downloading agent package from https://gbl.his.arc.azure.com/azcmagent/latest/AzureConnectedMachineAgent.msi to
C:\Users\Mallikah\AppData\Local\Temp\AzureConnectedMachineAgent.msi
VERBOSE: Installing agent package
```

You will be asked to confirm your identity by logging into your Microsoft Account.

Verify the Installation:

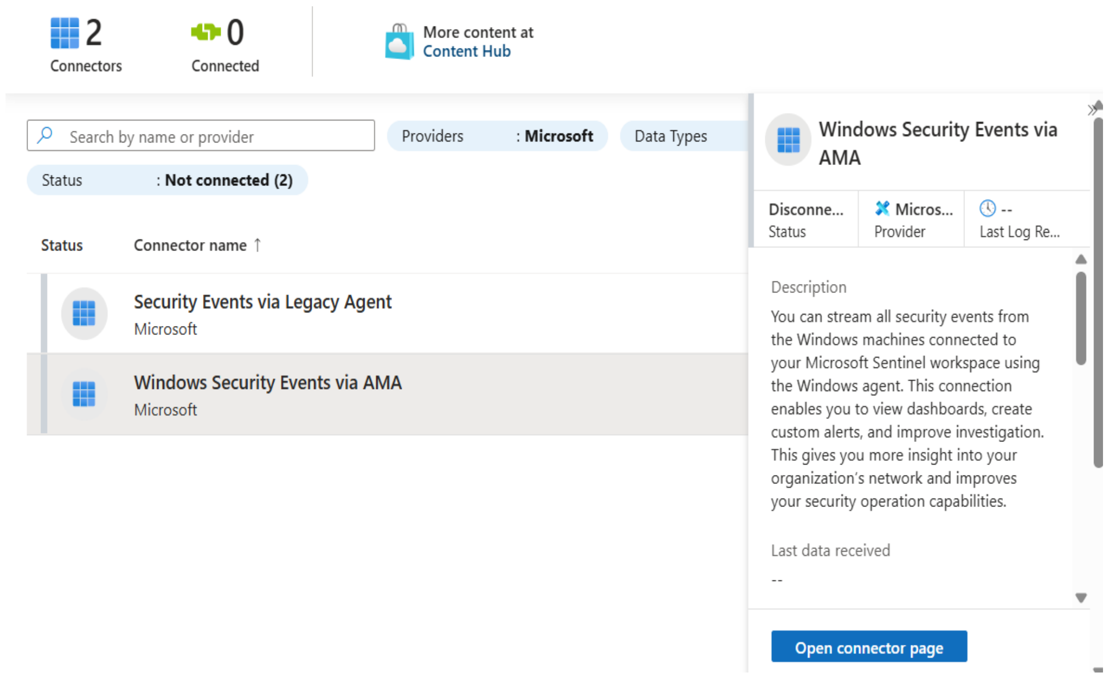
After running the script on your device, return to the Azure portal and verify that the device appears under the **All Azure Arc Resources** by refreshing the page.

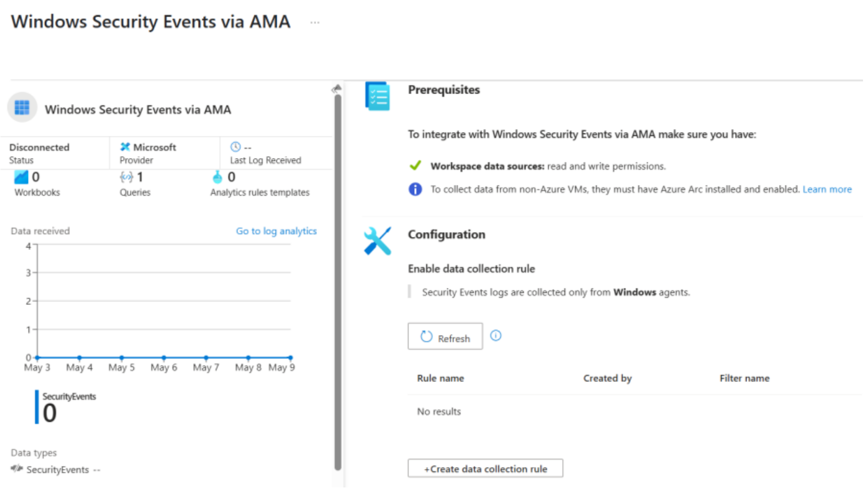


Step 7: Set Up a Data Collection Rule for the Azure Arc Device from Sentinel's Data Connector

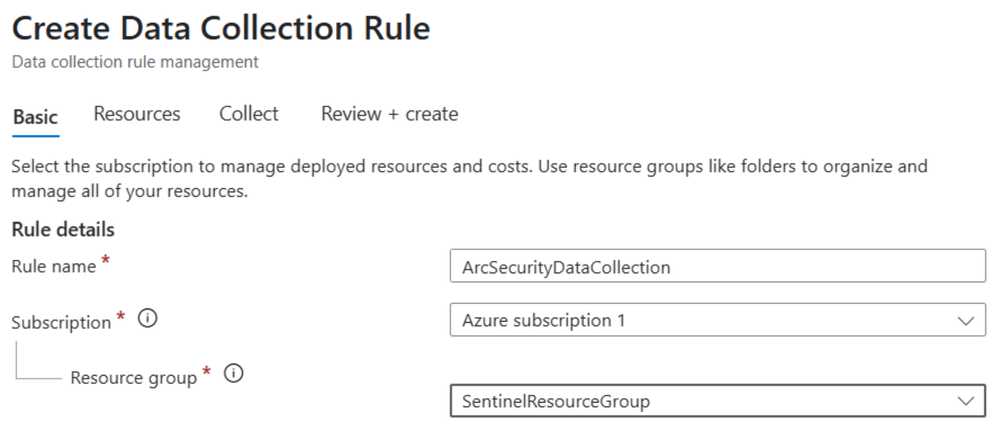
Now that your device is connected via **Azure Arc**, you need to ensure the logs from that device are collected. You'll set up a **Data Collection Rule (DCR)** from the **Windows Security Events** Data Connector in Sentinel.

1. **Go Back to Microsoft Sentinel**:
   - Navigate back to **Microsoft Sentinel**.
2. **Access the Windows Security Events Data Connector**:
   - Click on **Data connectors** from the Sentinel menu.
   - Search for **Windows Security Events** via AMA (Azure Monitor Agent) click on the '**Open Connector Page**' button followed by the '+**Create data collection** rule.

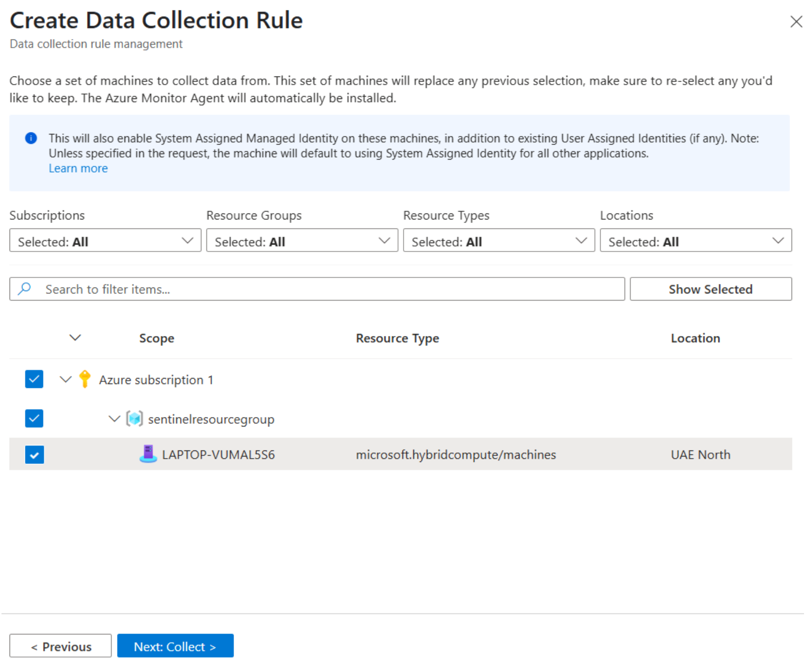




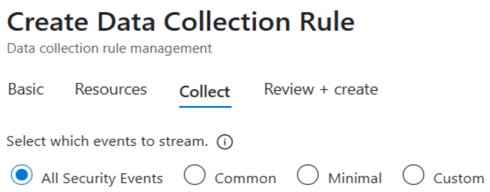
3. **Create Data Collection Rule (DCR)**:
   - You'll be prompted to create a **Data Collection Rule** (DCR).
   - Provide a **Name** for your rule (e.g., *ArcSecurityDataCollection*).
   - Select the **Subscription** and **Resource Group** (SentinelResourceGroup) where your Arc-connected device is located.
   - Select the **Region** that matches your Arc-connected device.



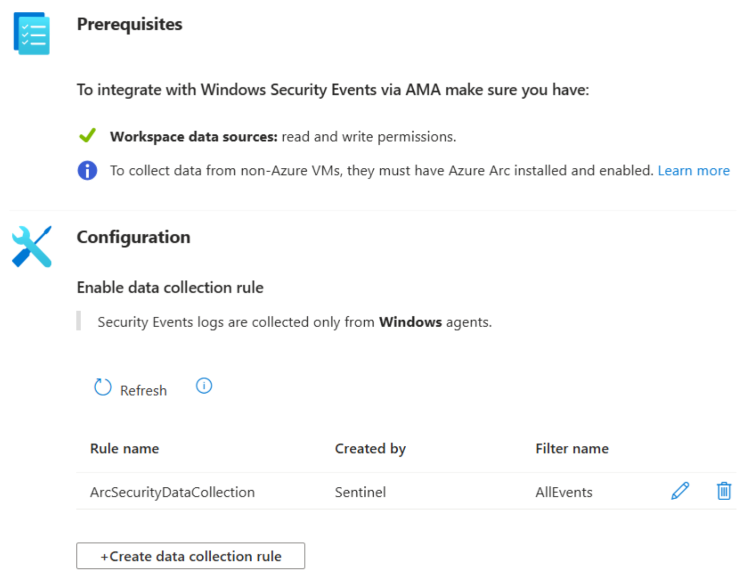
4. **Add Data Sources**:
   - In the **Data Sources** section, select **Windows Security Events** and other relevant event types that you want to collect from your Arc-connected device.
5. **Assign the Azure Arc-Connected Device**:
   - In the **Resources** section, select your **Azure Arc-connected device** from the list of available resources.
6. **Select Log Analytics Workspace**:
   - Ensure the **Destination** of the logs is set to your **Log Analytics Workspace** (*SentinelLogsWorkspace*).
7. **Review and Create**:
   - Review your settings, then click **Next: Collect** to finalize the Data Collection Rule.



Leave the collect rule management the same as “All security events”.



Then review and create your Data Collection Rule. You will see a confirmation of your rule creation in the Windows Security Events via AMA connector page.



You have successfully set up the Microsoft Sentinel collecting data from your device.

Conclusion

A Security Information and Event Management (SIEM) system is essential to the work of a SOC analyst. It empowers analysts to detect, investigate, and respond to security incidents with greater speed and accuracy. With features like automated alerts, customizable dashboards, and advanced analytics, SIEMs help pinpoint suspicious behaviour and uncover potential vulnerabilities. By automating routine tasks, they improve operational efficiency and free analysts to focus on critical threats. In today's fast-changing threat landscape, where attacks can emerge and evolve quickly, a SIEM is crucial to ensuring no threat goes unnoticed.